

# FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

## 1. Case Details

|                   |                                                       |
|-------------------|-------------------------------------------------------|
| Case ID           | CASE-7AC9B30F 7ac9b30f-9bbd-4f52-b0db-2952c145ead4    |
| Submitted by      | testreviewer L!                                       |
| Submission date   | 2026-10-07 06:44 UTC                                  |
| Documents         | 1    Types: School document                           |
| Case type         | School document                                       |
| Status / decision | Submitted / <b>Pending review</b>                     |
| Report generated  | 2026-10-07 06:47 UTC by testreviewer L! (Reviewer L1) |
| Report ID         | fc09df48-3771-4873-bae0-7346204e254e                  |

**Preliminary report.** Automated analysis had not finished when this report was generated. Still outstanding: Field validation of 'case11School educational document.pdf'; Issuer verification of 'case11School educational document.pdf'; Metadata forensics of 'case11School educational document.pdf'; Error level analysis of 'case11School educational document.pdf'; Copy move detection of 'case11School educational document.pdf'; Duplicate detection of 'case11School educational document.pdf'.

## 2. Executive Summary

**Not yet scored.** The risk engine has not produced an assessment for this case, so there is no overall tier.

**Why:** No risk tier yet — automated analysis is still in progress.

**Decision status:** Pending review.

**Exceptions:** none — every check that ran passed.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

### 3. Explainable Findings

The case has not been scored, so there are no triggered risk reasons yet.

## 4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

**Document 1 — case11School educational document.pdf** (School document · 1 passed, 0 flagged)

| Check                                                                                                                                  | Result | Details                            |
|----------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------|
| <b>Font consistency</b><br>Text set in a different font family from the text around it (text layer, or OCR font recognition on scans). | Pass   | No anomaly detected by this check. |

## 5. Exceptions

No exceptions — every check that ran passed.

## 6. Limitations and Assumptions

Please read this section before relying on anything in this report.

### Limitations of This Analysis

#### What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

#### Three kinds of finding

**Deterministic pixel and structure checks** — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

**Deterministic rule-based field checks** — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

**Model-judgment checks** — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

#### What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

#### Absence of a finding

**The absence of a flagged finding does not guarantee that a document is genuine.** It means only that no supported check detected an anomaly.

#### Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

| Check                                       | Status         | Coverage on this case                                       |
|---------------------------------------------|----------------|-------------------------------------------------------------|
| Metadata forensics                          | Did not run    | Did not run on any of the 1 document(s).                    |
| Font consistency                            | Ran            | Ran on 1 of 1 document(s); 0 flagged.                       |
| Error level analysis (ELA)                  | Did not run    | Did not run on any of the 1 document(s).                    |
| Copy-move detection                         | Did not run    | Did not run on any of the 1 document(s).                    |
| Visual inconsistency / AI-generation review | Did not run    | Did not run on any of the 1 document(s).                    |
| Signature / stamp detection                 | Did not run    | Did not run on any of the 1 document(s).                    |
| Duplicate detection                         | Did not run    | Did not run on any of the 1 document(s).                    |
| Field validation                            | Did not run    | Did not run on any of the 1 document(s).                    |
| Issuer verification                         | Did not run    | Did not run on any of the 1 document(s).                    |
| Signature comparison (advisory)             | Did not run    | Did not run — no reference signature was set for this case. |
| Cross-document consistency                  | Not applicable | Not applicable — this case has a single document.           |

#### Risk score and rule versions

This case had not been scored when the report was generated, so no risk-rule versions apply.

### Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: no field was marked uncertain on this case.
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.



## 7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

| Time                 | Event                                                          | Actor                         | Detail                                                             |
|----------------------|----------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------|
| 2026-10-07 06:44 UTC | Case created                                                   | testreviewer L! (Reviewer L1) |                                                                    |
| 2026-10-07 06:44 UTC | Document uploaded                                              | testreviewer L! (Reviewer L1) | case11School educational document.pdf; SHA-256 624ebf7b699285dd... |
| 2026-10-07 06:47 UTC | Font consistency check completed                               | System                        | case11School educational document.pdf                              |
| 2026-10-07 06:47 UTC | Text extraction, classification and field extraction completed | System                        | case11School educational document.pdf                              |

## 8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

### 8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

#### Document 1 — case11School educational document.pdf

| Field                   | Value                                                                | Confidence | Located |
|-------------------------|----------------------------------------------------------------------|------------|---------|
| Issuer                  | PINK & BLUE NURSERY Al Ain - UAE حضانة الوردى والأزرق                | 95%        | p.1     |
| Reference number        | MLC15112287877                                                       | 95%        | p.1     |
| Date                    | 2022-11-15                                                           | 95%        | p.1     |
| Amount                  | 10,300.00 AED                                                        | 95%        | p.1     |
| Subtotal                | 10,300.00 AED                                                        | 95%        | p.1     |
| Tax rate                | 5.00                                                                 | 90%        | p.2     |
| Tax amount              | 0.00 AED                                                             | 90%        | p.2     |
| Due date                | 15-Nov-2022                                                          | 95%        | p.1     |
| Bill to student details | Hamdan Khan Pink and Blue Al Khabeesi - Br1                          | 95%        | p.1     |
| Remarks description     | Monthly Payment Fee (Partial) * Note: Registration is Non-Refundable | 90%        | p.1     |

### 8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

| Document                                              | SHA-256                                                          | Verification |
|-------------------------------------------------------|------------------------------------------------------------------|--------------|
| case11School educational document.pdf<br>29,904 bytes | 624ebf7b699285dd3f499693075b4f4ac5f600fe671da3faaeb2b3c80baaf537 | VERIFIED     |

### 8.3 Technical parameters used by each check

#### Page rendering (input to ELA, copy-move, duplicate, visual review)

|                   |                                  |
|-------------------|----------------------------------|
| Render resolution | 200 DPI                          |
| Colour model      | RGB, converted to BGR for OpenCV |

#### Field locations (purple highlights)

|                 |                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------|
| Method          | extracted values matched back onto OCR word positions (Azure Document Intelligence Layout) |
| Repeated values | resolved by nearby keywords (“total”, “subtotal”, “VAT”, “date”...)                        |
| Accuracy        | approximate; a field that cannot be matched is reported without a highlight                |

## 9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

No check produced a highlight for this case, so no pages are rendered.